Expt. No: 04 Design a simple topology and configure with one Router, Two Switches and PC using CISCO Packets

1. Build the Topology:
   i) Add End Devices
   ii) Add Switches
   iii) Add a Router
   iv) Cable the devices
      • Connect PC0 to Switch

2. Configure the PCs:
   Configure each host with its unique static IPV4 address, subnet mask, and default gateway.
   (i) Click on PC0 and open its configuration panel.
   (ii) Go to Desktop tab and click on IP configuration.
   (iii) Select Static & fill in:
      • IPV4 address:
        192.168.10.10
      • Subnet Mask:
        255.255.255.0
      • Default Gateway:
        192.168.10.1
   (iv) Repeat this exact workflow for the remaining endpoints:
      • PC1 configuration:
        • IP address: 192.168.10.11
        • Subnet Mask: 255.255.255.0
        • Default Gateway: 192.168.10.1

3. Configure
- Router > enable
- Router # configure terminal
- Router (config) # interface GigabitEthernet 0/0/1
- Router (config-if) # ip address 192.168.
255.255.255.0
- Router (config-if) # no shutdown
- Router (config-if) # exit
- Router (config) # interface GigabitEthernet 0/0/0
- Router (config-if) # ip address 192.168.
255.255.255.0
- Router (config-if) # no shutdown
- Router (config-if) # exit
- Router (config) # do write memory.

4. Verify connectivity:
- click on PC0, go to the Desktop tab, and then command Prompt
- Run a command line ping to your local gateway to confirm the link: ping 192.168.10.
- Test cross-subnet routing across the network layer by pinging a cross-network ping to PC2: ping 192.168.20.10.

Result:

The network topology is operational. The first individual packet in inter-subnet ping may initially time out due to Address Resolution Protocol (ARP) discovery, but subsequent requests will return successful replies from 192.168.20.10 with zero packet loss.

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